6.2 Software Integrity

Control Type: Mandatory / Advisory for A4	Applies to architecture:	A1	A2	A3	A4	B
		●	●	●	●	

Control Definition

Control Objective: Ensure the software integrity of the Swift-related components and act upon results.

In-scope components:

- Swift connector
- GUI to the messaging and communication interface
- messaging interface
- communication interface
- Local RMA
- SNL
- New HSM (only for the checking of the firmware update)
- [Advisory: Customer (client) connector]

Risk Drivers:

- unauthorised system changes
- HSM management misused

Implementation Guidance

Control Statement:

A software integrity check is performed at regular intervals on messaging interface, communication interface, and other Swift-related components and results are considered for appropriate resolving actions. Origin and integrity of the software is ensured at download and at deployment time.

Control Context:

Software integrity checks provide a detective control against unexpected modification to operational software.

Implementation Guidelines:

The implementation guidelines are common methods to apply the relevant control. The guidelines are a helpful way to begin an assessment but should never be considered as an audit checklist as each user's implementation may vary. Therefore, in cases where some implementation guidelines elements are not present or partially covered, mitigations as well as particular environment specificities must be considered to properly assess the overall compliance adherence level (as per the suggested guidelines or as per the alternatives).

- Software integrity checks are conducted on in-scope components upon start-up, and additionally at least once per day.

  Options for implementation:
  - Integrated in the product
  - Third-party file integrity monitoring (FIM) tool
- Before applying a downloaded software that has either been developed by a third-party or in-house, operators should validate the legitimate external or internal source/site and perform, when technically possible, integrity checks such as checksum validation to support the change and release management process up to the deployment of the software.
- The resulting alerts are analysed for appropriate action and remediation in line with or as part of the control 6.4.

Optional Enhancements:

- An integrity check is performed in memory.
- An integrity check is performed at the operating system level.
- File Integrity Monitoring covers the products with integrated mechanisms.